Aman Shaikh F-02

Aim:Implement Random Forest and XGboost classifiers and compare their performance with basic decision tree classifiers.

```
import pandas as pd
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score,f1_score, confusion_matrix
```

```
data = load_wine()
X = data.data
y = data.target
X_train, X_test, y_train, y_test = train_test_split(
     X, y, test_size = 0.3, random_state =42)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
#Decision tree
dt = DecisionTreeClassifier()
dt.fit(X_train, y_train)
y_pred_dt = dt.predict(X_test)

# Random Forest
rf = RandomForestClassifier(n_estimators = 200, random_state = 42)
rf.fit(X_train, y_train)
y_pred_rf = rf.predict(X_test)

# XGBoost
xgb = XGBClassifier(use_label_encoder = False, eval_metric = 'mlogloss')
xgb.fit(X_train, y_train)
y_pred_xgb = xgb.predict(X_test)
```

```
/usr/local/lib/python3.12/dist-packages/xgboost/training.py:200: UserWarning: [04:31:51] WARNING: /__w/xgboost/xg
Parameters: { "use_label_encoder" } are not used.

  bst.update(dtrain, iteration=i, fobj=obj)
```

```
# 7. Evaluation Function
print("\ndecision Tree")
print("Accuracy :", accuracy_score(y_test, y_pred_dt))
print("Precision :", precision_score(y_test, y_pred_dt, average = 'macro'))
print("Recall :", recall_score(y_test, y_pred_dt, average = 'macro'))
print("F1 Score :", f1_score(y_test, y_pred_dt, average = 'macro'))
print("\nRandom Forest")
print("Accuracy :", accuracy_score(y_test, y_pred_rf))
print("Precision :", precision_score(y_test, y_pred_rf, average = 'macro'))
print("Recall :", recall_score(y_test, y_pred_rf, average = 'macro'))
print("F1 Score :", f1_score(y_test, y_pred_rf, average = 'macro'))

print("\nXGBoost")
print("Accuracy :", accuracy_score(y_test, y_pred_xgb))
print("Precision :", precision_score(y_test, y_pred_xgb, average = 'macro'))
print("Recall :", recall_score(y_test, y_pred_xgb, average = 'macro'))
print("F1 Score :", f1_score(y_test, y_pred_xgb, average = 'macro'))
```

```

decision Tree
Accuracy : 0.9629629629629629
Precision : 0.9710144927536232
Recall : 0.9586466165413533
F1 Score : 0.9634937968271302
```

```
Random Forest
Accuracy : 1.0
Precision : 1.0
Recall : 1.0
F1 Score : 1.0

XGBoost
Accuracy : 0.9814814814814815
Precision : 0.9848484848484849
Recall : 0.9761904761904763
F1 Score : 0.9799023830031581
```